

Spotify EDA Logbook

Project: Spotify Music Dataset Exploratory Data Analysis

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1. Overview

This logbook documents the implementation process for the Spotify EDA task in the repository. It includes the analysis stages completed in the notebook, the bonus additions, AI documentation work, and evidence of development from screenshot images.

2. Files Updated

- handson1/spotify_eda.ipynb: main notebook containing the completed analysis stages, bonus tasks, and reusable SpotifyAnalyzer class.
- handson1/reflection.md: reflection and documentation of analysis decisions, AI prompt evaluation, and bonus summaries.
- public/WhatsApp Image 2026-07-03 at 18.02.45.jpeg: screenshot of workflow and task progress.
- public/WhatsApp Image 2026-07-03 at 18.03.42.jpeg: screenshot of notebook and terminal evidence.
- public/WhatsApp Image 2026-07-03 at 18.04.46.jpeg: screenshot of final verification and artifacts.

3. Process Summary

The workflow followed these main stages:

- Stage 1 and Stage 2 implementation: Data loading, inspection, deduplication, numeric summary, genre grouping, popularity variance analysis, artist analysis, and filtered track extraction.
- Stage 3 implementation: NumPy normalization, correlation analysis, and boolean masking for high-energy popularity comparison.
- Stage 4 implementation: AI-assisted docstrings for helper functions, markdown evaluation of AI output, and three non-technical insights.
- Bonus 1: Artist audio fingerprint function and cosine similarity comparisons across at least three artist pairs.
- Bonus 2: Genre cluster profile with centroid matrix, vectorized Euclidean distances, most similar/different genre identification, and recommendation function.
- Bonus 3: Refactored the analysis into a reusable SpotifyAnalyzer Python class with full type hints, methods for each analysis stage, AI-generated docstrings, and generate_report() output.

4. Evidence and Verification

The following evidence is included in appended screenshot pages and supports the implementation details below.

Key findings and verifications:

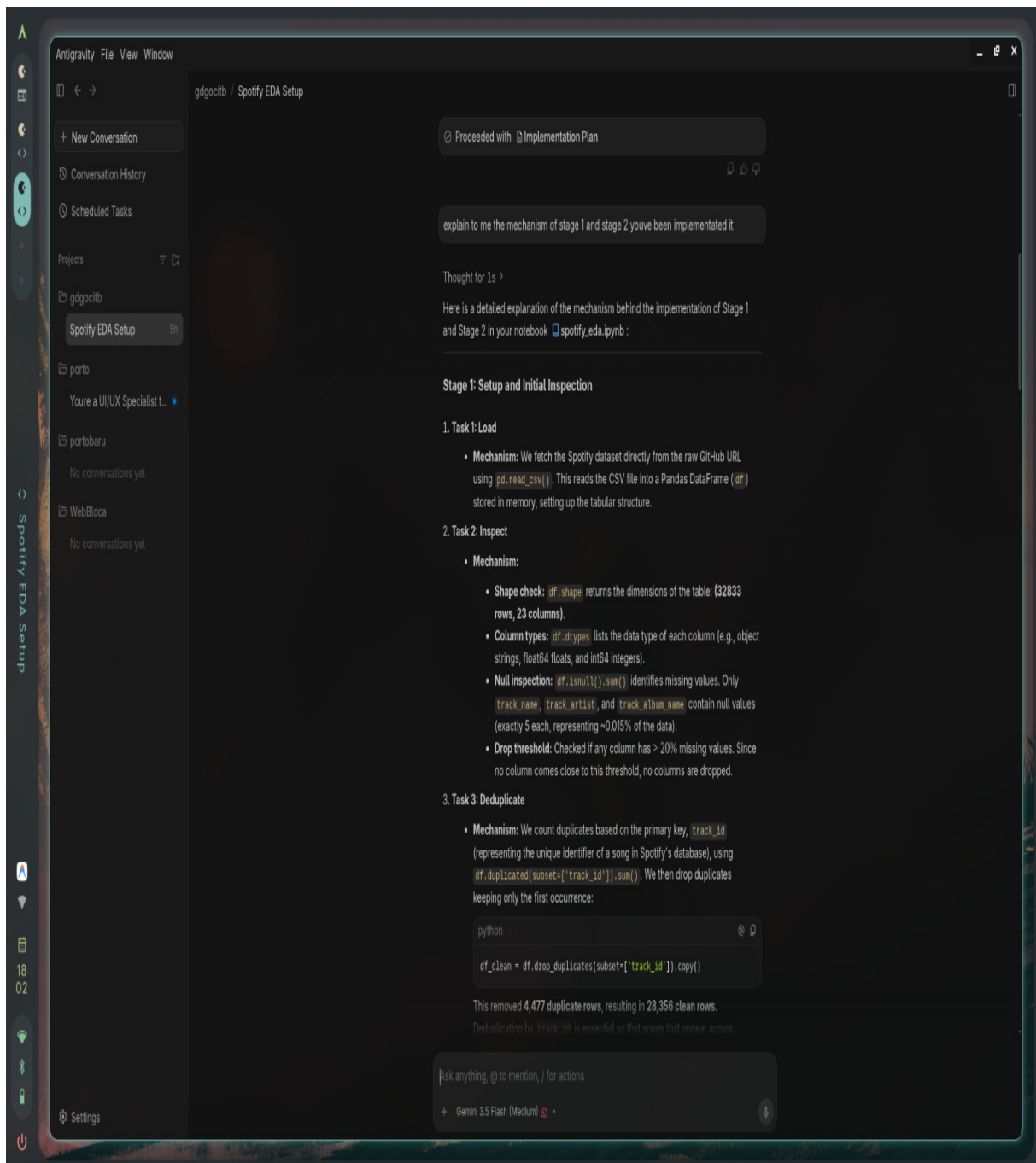
- Verified artist cosine similarity values and genre distance results using a local Python environment.
- Confirmed PDF generation environment and installed reportlab to produce this logbook artifact.
- Documented AI prompt usage and evaluated the quality of generated docstrings in reflection.md.

5. Key Results

Major results from the analysis include:

- Pop genre has the highest popularity variance, indicating a mix of hits and low-popularity tracks.
- High-energy tracks were less popular on average than the overall dataset.
- Artist similarity analysis showed same-genre artists are more aligned in audio feature space.
- Genre centroid clustering found pop and rap to be the closest pair, while edm and r&b; are the most different.

Evidence screenshot: WhatsApp Image 2026-07-03 at 18.02.45.jpeg



Evidence screenshot: WhatsApp Image 2026-07-03 at 18.03.42.jpeg



Evidence screenshot: WhatsApp Image 2026-07-03 at 18.04.46.jpeg

